

Operations Research

Problem Set 2 – Detailed Solutions

Conventions as in Lectures 2–4: equational form with slack variables $s_i \geq 0$, equation-style tableaux $x_i = p_i + \sum_{j \in N} q_{ij}x_j$, $v = v_0 + \sum_{j \in N} r_j x_j$, reduced costs r_j , and the primal–dual pair (P): $\max c^\top x$, $Ax \leq b$, $x \geq 0$ / (D): $\min b^\top y$, $A^\top y \geq c$, $y \geq 0$.

Problem 1 – Herb blends

Solution 1(a)

Let x_1 and x_2 be the quantities of blend A and blend B produced per day, in units of 100 kg. The profit per unit is selling price minus variable cost: $900 - 400 = 500$ MAD for A and $700 - 300 = 400$ MAD for B, i.e. 5 and 4 hundred MAD. The processing and packaging hours give one inequality each. In inequality form,

$$\begin{array}{ll} \max & 5x_1 + 4x_2 \\ \text{s.t.} & 2x_1 + x_2 \leq 10 \quad (\text{processing hours}), \\ & x_1 + 2x_2 \leq 8 \quad (\text{packaging hours}), \\ & x_1, x_2 \geq 0. \end{array}$$

The objective $v = 5x_1 + 4x_2$ is the daily profit in hundreds of MAD.

Solution 1(b)

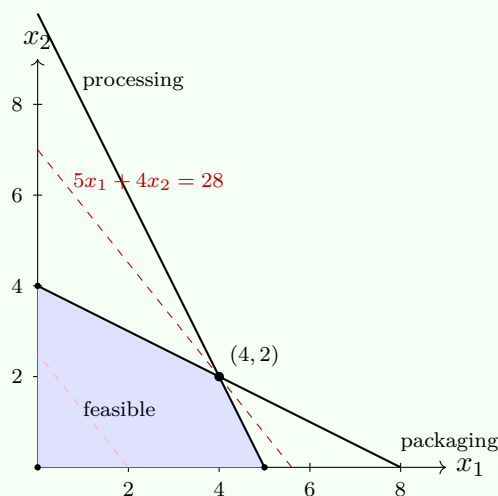
Each inequality is a half-plane. The processing line $2x_1 + x_2 = 10$ meets the axes at $(5, 0)$ and $(0, 10)$; the packaging line $x_1 + 2x_2 = 8$ at $(8, 0)$ and $(0, 4)$. They intersect where

$$2x_1 + x_2 = 10, \quad x_1 + 2x_2 = 8 \implies x_2 = 10 - 2x_1, \quad x_1 + 20 - 4x_1 = 8 \implies x_1 = 4, \quad x_2 = 2.$$

The feasible region is the quadrilateral with vertices

$$(0, 0), \quad (5, 0), \quad (4, 2), \quad (0, 4).$$

(The point $(8, 0)$ violates processing, $8 \cdot 2 = 16 > 10$; the point $(0, 10)$ violates packaging, $20 > 8$.)



The level lines of the objective, $5x_1 + 4x_2 = \text{const}$, have slope $-5/4$. Sliding one in the direction of increasing profit, the last feasible point it touches is the vertex $(4, 2)$. Checking all

four vertices confirms this:

$$v(0, 0) = 0, \quad v(5, 0) = 25, \quad v(4, 2) = 28, \quad v(0, 4) = 16.$$

Hence $x_1^* = 4$, $x_2^* = 2$, $v^* = 28$.

Uniqueness. The optimal level line $5x_1 + 4x_2 = 28$ has slope $-5/4$, while the two edges meeting at $(4, 2)$ have slopes -2 (processing) and $-1/2$ (packaging). Since $-5/4$ equals neither, the level line is not parallel to any edge, so it meets the feasible region in the single point $(4, 2)$. The optimum is unique.

Solution 1(c)

Introduce unused processing hours $s_1 \geq 0$ and unused packaging hours $s_2 \geq 0$:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 + s_1 = 10, \\ & x_1 + 2x_2 + s_2 = 8, \\ & x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

With variable order (x_1, x_2, s_1, s_2) ,

$$A = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 10 \\ 8 \end{pmatrix}, \quad c = \begin{pmatrix} 5 \\ 4 \\ 0 \\ 0 \end{pmatrix}.$$

There are $n = 4$ variables and $m = 2$ equations. The last two columns form I_2 , so the two rows are independent and $\text{rank}(A) = 2 = m$.

Solution 1(d)

A basis is a set B of $m = 2$ column indices with A_B invertible. The four columns $(2, 1)^\top, (1, 2)^\top, (1, 0)^\top, (0, 1)^\top$ are nonzero and no two are proportional, so every one of the $\binom{4}{2} = 6$ pairs is independent: all six selections are bases. For each basis, set the two nonbasic variables to zero and solve $A_B z_B = b$.

- $B = \{x_1, x_2\}$: $2x_1 + x_2 = 10$, $x_1 + 2x_2 = 8$, hence $(x_1, x_2) = (4, 2)$ as in (b).
- $B = \{x_1, s_1\}$: $x_2 = s_2 = 0$ gives $x_1 = 8$ from the second equation, then $s_1 = 10 - 16 = -6$.
- $B = \{x_1, s_2\}$: $x_2 = s_1 = 0$ gives $x_1 = 5$, then $s_2 = 8 - 5 = 3$.
- $B = \{x_2, s_1\}$: $x_1 = s_2 = 0$ gives $x_2 = 4$, then $s_1 = 10 - 4 = 6$.
- $B = \{x_2, s_2\}$: $x_1 = s_1 = 0$ gives $x_2 = 10$, then $s_2 = 8 - 20 = -12$.
- $B = \{s_1, s_2\}$: $x_1 = x_2 = 0$ gives $(s_1, s_2) = (10, 8)$.

Basis	x_1	x_2	s_1	s_2	Feasible?	v (10^2 MAD)
$\{x_1, x_2\}$	4	2	0	0	Yes	28
$\{x_1, s_1\}$	8	0	-6	0	No	—
$\{x_1, s_2\}$	5	0	0	3	Yes	25
$\{x_2, s_1\}$	0	4	6	0	Yes	16
$\{x_2, s_2\}$	0	10	0	-12	No	—
$\{s_1, s_2\}$	0	0	10	8	Yes	0

Four of the six basic solutions are feasible; they are the four BFSs, and they project exactly onto the four vertices of (b). The LP is feasible (the origin) and its objective is bounded above (for instance $v = 5x_1 + 4x_2 \leq 5(2x_1 + x_2) \leq 50$), so by the fundamental existence theorem of Lecture 2 an optimal BFS exists. Comparing the four values, the optimum is the BFS of $\{x_1, x_2\}$:

$$(x_1^*, x_2^*, s_1^*, s_2^*) = (4, 2, 0, 0), \quad v^* = 28.$$

Solution 1(e)

Since $b \geq 0$, the slack basis $\{s_1, s_2\}$ is feasible and gives the starting tableau. Throughout, the entering variable is one with a positive reduced cost (we take the largest; any positive one is permitted), and the leaving variable is given by the smallest row limit.

Tableau 0 (basic s_1, s_2 ; nonbasic x_1, x_2).

$$s_1 = 10 - 2x_1 - x_2$$

$$s_2 = 8 - x_1 - 2x_2$$

$$v = 0 + 5x_1 + 4x_2$$

BFS $(x_1, x_2, s_1, s_2) = (0, 0, 10, 8)$, $v = 0$. Both reduced costs are positive; the larger is $r_1 = 5$, so x_1 enters. Keep $x_2 = 0$. Row limits: $s_1 = 10 - 2x_1 \geq 0$ gives $x_1 \leq 5$; $s_2 = 8 - x_1 \geq 0$ gives $x_1 \leq 8$. The smallest limit is 5, from the s_1 row: s_1 leaves.

Pivot 1. Solve the leaving row for the entering variable: $s_1 = 10 - 2x_1 - x_2 \iff x_1 = 5 - \frac{1}{2}x_2 - \frac{1}{2}s_1$. Substitute into the other rows and the objective:

$$s_2 = 8 - (5 - \frac{1}{2}x_2 - \frac{1}{2}s_1) - 2x_2 = 3 - \frac{3}{2}x_2 + \frac{1}{2}s_1,$$

$$v = 5(5 - \frac{1}{2}x_2 - \frac{1}{2}s_1) + 4x_2 = 25 + \frac{3}{2}x_2 - \frac{5}{2}s_1.$$

Tableau 1 (basic x_1, s_2 ; nonbasic x_2, s_1).

$$x_1 = 5 - \frac{1}{2}x_2 - \frac{1}{2}s_1$$

$$s_2 = 3 - \frac{3}{2}x_2 + \frac{1}{2}s_1$$

$$v = 25 + \frac{3}{2}x_2 - \frac{5}{2}s_1$$

BFS $(5, 0, 0, 3)$, $v = 25$: the vertex $(5, 0)$. The only positive reduced cost is $r_{x_2} = \frac{3}{2}$, so x_2 enters. Keep $s_1 = 0$. Row limits: $x_1 = 5 - \frac{1}{2}x_2 \geq 0$ gives $x_2 \leq 10$; $s_2 = 3 - \frac{3}{2}x_2 \geq 0$ gives $x_2 \leq 2$. The smallest is 2: s_2 leaves.

Pivot 2. $s_2 = 3 - \frac{3}{2}x_2 + \frac{1}{2}s_1 \iff x_2 = 2 + \frac{1}{3}s_1 - \frac{2}{3}s_2$. Substitute:

$$x_1 = 5 - \frac{1}{2}(2 + \frac{1}{3}s_1 - \frac{2}{3}s_2) - \frac{1}{2}s_1 = 4 - \frac{2}{3}s_1 + \frac{1}{3}s_2,$$

$$v = 25 + \frac{3}{2}(2 + \frac{1}{3}s_1 - \frac{2}{3}s_2) - \frac{5}{2}s_1 = 28 - 2s_1 - s_2.$$

Tableau 2 (basic x_1, x_2 ; nonbasic s_1, s_2).

$$x_1 = 4 - \frac{2}{3}s_1 + \frac{1}{3}s_2$$

$$x_2 = 2 + \frac{1}{3}s_1 - \frac{2}{3}s_2$$

$$v = 28 - 2s_1 - s_2$$

BFS $(4, 2, 0, 0)$, $v = 28$.

Stopping decision. Both reduced costs are negative, $r_{s_1} = -2$ and $r_{s_2} = -1$. The objective row is an identity valid for every feasible solution, and $s_1, s_2 \geq 0$, so

$$v = 28 - 2s_1 - s_2 \leq 28 \quad \text{for every feasible point.}$$

The current BFS attains 28; it is therefore optimal, and the algorithm stops. Moreover $v = 28$ forces $s_1 = s_2 = 0$, which through the two basic rows forces $x_1 = 4$, $x_2 = 2$: the optimum is unique, in agreement with (b).

The path taken was $(0, 0) \rightarrow (5, 0) \rightarrow (4, 2)$ along two edges of the polygon, visiting three of the four BFSs of (d).

Solution 1(f)

In original units, the cooperative should produce

400 kg of blend A and 200 kg of blend B per day,

for a daily profit of $28 \times 100 = 2800$ MAD. Resource use: processing $2(4) + 2 = 10$ hours of 10, packaging $4 + 2(2) = 8$ hours of 8. Both slacks are zero: there are no unused resources; both constraints are binding.

The enumeration in (d) lists only the *basic* feasible solutions, which are finitely many (at most $\binom{4}{2} = 6$). The feasible region is a polygon with infinitely many points, for instance $(1, 1)$ or $(2, 1)$, none of which is basic. Enumeration is nevertheless sufficient because the fundamental existence theorem guarantees that, for a feasible LP with bounded objective, some optimal solution is a BFS; so the best BFS is an optimal solution of the whole LP.

Problem 2 – Computing centre

Solution 2(a)

Let $x_1, x_2, x_3 \geq 0$ be the amounts (in GB) of type 1, 2 and 3 work processed per day. Score and resource use are proportional to the amounts, so

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 + 5x_3 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4 \quad (\text{CPU hours}), \\ & x_2 + x_3 \leq 3 \quad (\text{GPU hours}), \\ & x_1 + x_3 \leq 3 \quad (\text{outbound GB}), \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Solution 2(b)

Add slacks $s_1, s_2, s_3 \geq 0$ for unused CPU hours, GPU hours and transfer capacity:

$$\begin{aligned} x_1 + x_2 + s_1 &= 4, \\ x_2 + x_3 + s_2 &= 3, \\ x_1 + x_3 + s_3 &= 3, \end{aligned} \quad x_1, x_2, x_3, s_1, s_2, s_3 \geq 0,$$

so $n = 6$, $m = 3$, and the slack columns form I_3 , giving rank $A = 3$. Because $b = (4, 3, 3) \geq 0$, the slack basis $B = \{s_1, s_2, s_3\}$ is feasible:

$$(x_1, x_2, x_3, s_1, s_2, s_3) = (0, 0, 0, 4, 3, 3), \quad v = 0.$$

Its tableau is

Tableau 0 (basic s_1, s_2, s_3 ; nonbasic x_1, x_2, x_3).

$$\begin{aligned} s_1 &= 4 - x_1 - x_2 \\ s_2 &= 3 \quad \quad - x_2 - x_3 \\ s_3 &= 3 - x_1 \quad \quad - x_3 \\ v &= 0 + 5x_1 + 4x_2 + 5x_3 \end{aligned}$$

Solution 2(c)

Tableau 0. Reduced costs $(5, 4, 5)$, all positive. The largest value 5 is attained by x_1 and x_3 ; we break the tie by the smallest index and let x_1 enter. (Choosing x_3 is equally valid and leads to the same optimum by a different path.) Keep $x_2 = x_3 = 0$. Row limits: $s_1 = 4 - x_1 \geq 0$ gives $x_1 \leq 4$; s_2 does not involve x_1 ; $s_3 = 3 - x_1 \geq 0$ gives $x_1 \leq 3$. Smallest limit 3: s_3 leaves.

Pivot 1. $s_3 = 3 - x_1 - x_3 \iff x_1 = 3 - x_3 - s_3$. Substitute:

$$\begin{aligned} s_1 &= 4 - (3 - x_3 - s_3) - x_2 = 1 - x_2 + x_3 + s_3, \\ v &= 5(3 - x_3 - s_3) + 4x_2 + 5x_3 = 15 + 4x_2 + 0x_3 - 5s_3. \end{aligned}$$

Tableau 1 (basic x_1, s_1, s_2 ; nonbasic x_2, x_3, s_3).

$$\begin{aligned} x_1 &= 3 \quad \quad - x_3 - s_3 \\ s_1 &= 1 - x_2 + x_3 + s_3 \\ s_2 &= 3 - x_2 - x_3 \\ v &= 15 + 4x_2 + 0x_3 - 5s_3 \end{aligned}$$

BFS $(3, 0, 0, 1, 3, 0)$, $v = 15$. The reduced cost of x_3 has become 0: processing type 3 instead of type 1 would, at this point, change nothing, because both earn 5 and compete for the saturated transfer capacity. The only positive reduced cost is $r_{x_2} = 4$: x_2 enters. Keep $x_3 = s_3 = 0$. Row limits: x_1 does not involve x_2 ; $s_1 = 1 - x_2 \geq 0$ gives $x_2 \leq 1$; $s_2 = 3 - x_2 \geq 0$ gives $x_2 \leq 3$. Smallest limit 1: s_1 leaves.

Pivot 2. $s_1 = 1 - x_2 + x_3 + s_3 \iff x_2 = 1 + x_3 + s_3 - s_1$. Substitute:

$$\begin{aligned} s_2 &= 3 - (1 + x_3 + s_3 - s_1) - x_3 = 2 - 2x_3 + s_1 - s_3, \\ v &= 15 + 4(1 + x_3 + s_3 - s_1) - 5s_3 = 19 + 4x_3 - 4s_1 - s_3. \end{aligned}$$

Tableau 2 (basic x_1, x_2, s_2 ; nonbasic x_3, s_1, s_3).

$$\begin{aligned} x_1 &= 3 - x_3 \quad \quad - s_3 \\ x_2 &= 1 + x_3 - s_1 + s_3 \\ s_2 &= 2 - 2x_3 + s_1 - s_3 \\ v &= 19 + 4x_3 - 4s_1 - s_3 \end{aligned}$$

BFS $(3, 1, 0, 0, 2, 0)$, $v = 19$. The only positive reduced cost is $r_{x_3} = 4$: x_3 enters. Keep $s_1 = s_3 = 0$. Row limits: $x_1 = 3 - x_3 \geq 0$ gives $x_3 \leq 3$; $x_2 = 1 + x_3$ increases, no limit; $s_2 = 2 - 2x_3 \geq 0$ gives $x_3 \leq 1$. Smallest limit 1: s_2 leaves.

Pivot 3. $s_2 = 2 - 2x_3 + s_1 - s_3 \iff x_3 = 1 + \frac{1}{2}s_1 - \frac{1}{2}s_2 - \frac{1}{2}s_3$. Substitute:

$$\begin{aligned} x_1 &= 3 - (1 + \frac{1}{2}s_1 - \frac{1}{2}s_2 - \frac{1}{2}s_3) - s_3 = 2 - \frac{1}{2}s_1 + \frac{1}{2}s_2 - \frac{1}{2}s_3, \\ x_2 &= 1 + (1 + \frac{1}{2}s_1 - \frac{1}{2}s_2 - \frac{1}{2}s_3) - s_1 + s_3 = 2 - \frac{1}{2}s_1 - \frac{1}{2}s_2 + \frac{1}{2}s_3, \\ v &= 19 + 4(1 + \frac{1}{2}s_1 - \frac{1}{2}s_2 - \frac{1}{2}s_3) - 4s_1 - s_3 = 23 - 2s_1 - 2s_2 - 3s_3. \end{aligned}$$

Tableau 3 (basic x_1, x_2, x_3 ; nonbasic s_1, s_2, s_3).

$$\begin{aligned}x_1 &= 2 - \frac{1}{2}s_1 + \frac{1}{2}s_2 - \frac{1}{2}s_3 \\x_2 &= 2 - \frac{1}{2}s_1 - \frac{1}{2}s_2 + \frac{1}{2}s_3 \\x_3 &= 1 + \frac{1}{2}s_1 - \frac{1}{2}s_2 - \frac{1}{2}s_3 \\v &= 23 - 2s_1 - 2s_2 - 3s_3\end{aligned}$$

BFS $(2, 2, 1, 0, 0, 0)$, $v = 23$. All reduced costs are negative: the algorithm stops.

Solution 2(d)

The optimal plan processes

$$x_1^* = 2 \text{ GB of type 1}, \quad x_2^* = 2 \text{ GB of type 2}, \quad x_3^* = 1 \text{ GB of type 3},$$

for a daily score of $5(2) + 4(2) + 5(1) = 23$ points. In the original model,

$$\text{CPU: } 2 + 2 = 4 \leq 4, \quad \text{GPU: } 2 + 1 = 3 \leq 3, \quad \text{transfer: } 2 + 1 = 3 \leq 3.$$

All three constraints are satisfied, each with equality.

Solution 2(e)

The objective row of the final tableau, $v = 23 - 2s_1 - 2s_2 - 3s_3$, is an identity that every feasible solution satisfies, because it was obtained from the original equations by substitutions that preserve the solution set. Since every feasible solution has $s_1, s_2, s_3 \geq 0$,

$$v \leq 23 \quad \text{for every feasible solution,}$$

and the BFS $(2, 2, 1, 0, 0, 0)$ attains $v = 23$. It is therefore optimal. Since the three reduced costs are strictly negative, any feasible solution with $v = 23$ must have $s_1 = s_2 = s_3 = 0$, and the basic rows then give $x = (2, 2, 1)$: the optimum is unique.

Unused resources: $s_1^* = s_2^* = s_3^* = 0$, so no resource is left over; CPU, GPU and transfer capacity are all fully used. (Equivalently, the reduced costs $-2, -2, -3$ are the marginal values of one more hour of CPU, one more hour of GPU and one more GB of transfer: each resource is scarce.)

Problem 3 – Project crashing

Solution 3(a)

Let $x_A, x_B, x_C \geq 0$ be the number of days by which A, B and C are shortened, each at most 2. The shortened durations are $6 - x_A$, $5 - x_B$, $4 - x_C$. C starts when both A and B have finished, at time $\max(6 - x_A, 5 - x_B)$, so the project finishes at

$$\max(6 - x_A, 5 - x_B) + (4 - x_C) \leq 8.$$

A maximum is at most $8 - (4 - x_C)$ exactly when each of its arguments is, so this single constraint is equivalent to the two linear inequalities

$$(6 - x_A) + (4 - x_C) \leq 8 \iff x_A + x_C \geq 2, \quad (5 - x_B) + (4 - x_C) \leq 8 \iff x_B + x_C \geq 1.$$

The cost in thousands of MAD is $4x_A + 3x_B + 5x_C$. The LP is

$$\begin{aligned} \min \quad & 4x_A + 3x_B + 5x_C \\ \text{s.t.} \quad & x_A + x_C \geq 2 \quad (\text{path A-C}), \\ & x_B + x_C \geq 1 \quad (\text{path B-C}), \\ & x_A \leq 2, \\ & x_B \leq 2, \\ & x_C \leq 2, \\ & x_A, x_B, x_C \geq 0. \end{aligned}$$

To dualise with the rules of Lecture 4 we write all five constraints in the same direction. Multiplying the three upper bounds by -1 , the primal is $\min c^\top x$ subject to $Ax \geq b$, $x \geq 0$ with

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 1 \\ -2 \\ -2 \\ -2 \end{pmatrix}, \quad c = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}.$$

Solution 3(b)

The primal is a *minimisation*, so a dual solution must certify a *lower* bound on the cost. Multiply the five $\geq$ constraints by weights $y_1, \dots, y_5 \geq 0$ (nonnegative weights preserve $\geq$) and add:

$$(y_1 - y_3)x_A + (y_2 - y_4)x_B + (y_1 + y_2 - y_5)x_C \geq 2y_1 + y_2 - 2y_3 - 2y_4 - 2y_5.$$

Because $x_A, x_B, x_C \geq 0$, the left side is at most the cost $4x_A + 3x_B + 5x_C$ provided each coefficient is at most the corresponding cost. The best lower bound of this form is the dual

$$\begin{aligned} \max \quad & 2y_1 + y_2 - 2y_3 - 2y_4 - 2y_5 \\ \text{s.t.} \quad & y_1 - y_3 \leq 4 \quad (x_A), \\ & y_2 - y_4 \leq 3 \quad (x_B), \\ & y_1 + y_2 - y_5 \leq 5 \quad (x_C), \\ & y_1, y_2, y_3, y_4, y_5 \geq 0. \end{aligned}$$

In matrix form: $\max b^\top y$ s.t. $A^\top y \leq c$, $y \geq 0$.

Domains and associated constraints. Each dual variable corresponds to one primal constraint; each has domain $y_i \geq 0$ because every primal constraint is a $\geq$ inequality in a minimisation (the mirror image of the Lecture 4 rule “ $\leq$ row in a maximisation gives $y_i \geq 0$ ”):

Dual variable	Domain	Associated primal constraint	Meaning
y_1	≥ 0	$x_A + x_C \geq 2$	price (10^3 MAD/day) of the A-C deadline
y_2	≥ 0	$x_B + x_C \geq 1$	price of the B-C deadline
y_3	≥ 0	$-x_A \geq -2$, i.e. $x_A \leq 2$	price of the 2-day cap on A
y_4	≥ 0	$-x_B \geq -2$, i.e. $x_B \leq 2$	price of the 2-day cap on B
y_5	≥ 0	$-x_C \geq -2$, i.e. $x_C \leq 2$	price of the 2-day cap on C

Each dual constraint corresponds to one primal variable (x_A, x_B, x_C in that order), and is a $\leq$ inequality because that variable is nonnegative.

Solution 3(c)

Feasibility of the plan $(x_A, x_B, x_C) = (2, 1, 0)$. Check each constraint: $x_A + x_C = 2 \geq 2$ (tight); $x_B + x_C = 1 \geq 1$ (tight); $x_A = 2 \leq 2$ (tight); $x_B = 1 \leq 2$; $x_C = 0 \leq 2$; all nonnegative. The plan is feasible. Its cost is

$$4(2) + 3(1) + 5(0) = 11 \quad \text{thousand MAD, i.e. 11 000 MAD.}$$

(A finishes on day 4, B on day 4, C runs from day 4 to day 8.)

A dual certificate of the bound 8 000 MAD. Take $y = (4, 0, 0, 0, 0)$. Dual feasibility: $y_1 - y_3 = 4 \leq 4$; $y_2 - y_4 = 0 \leq 3$; $y_1 + y_2 - y_5 = 4 \leq 5$; $y \geq 0$. Its objective is $2(4) = 8$. By weak duality every feasible reduction plan costs at least 8 thousand MAD. In words: multiplying the constraint $x_A + x_C \geq 2$ by 4 gives $4x_A + 4x_C \geq 8$, and since $4x_A + 4x_C \leq 4x_A + 3x_B + 5x_C$ for $x \geq 0$, the cost is at least 8.

Interval. The feasible plan gives an upper bound and the dual solution a lower bound on the minimum cost C^* :

$$8\,000 \leq C^* \leq 11\,000 \text{ MAD.}$$

The largest possible saving relative to the proposed plan, using only these bounds, is $11\,000 - 8\,000 = 3\,000$ MAD. (The true saving may be smaller; the bounds only limit it.)

Solution 3(d)

Primal candidate. Shortening C helps both paths at once, so intuitively we trade C against A. Put $x_C = t \in [0, 2]$; then the cheapest way to satisfy the two deadline constraints is $x_A = 2 - t$ and $x_B = \max(1 - t, 0)$ (any larger values only add cost). The cost is

$$C(t) = 4(2 - t) + 3\max(1 - t, 0) + 5t = \begin{cases} 11 - 2t, & 0 \leq t \leq 1, \\ 8 + t, & 1 \leq t \leq 2, \end{cases}$$

which decreases until $t = 1$ and increases afterwards. The candidate is

$$x^* = (x_A, x_B, x_C) = (1, 0, 1), \quad \text{cost } 4 + 0 + 5 = 9.$$

It is feasible: $x_A + x_C = 2$, $x_B + x_C = 1$, all bounds respected.

Dual candidate via complementary slackness. Primal slacks at x^* : constraint 1 tight, constraint 2 tight, $x_A = 1 < 2$ (slack 1), $x_B = 0 < 2$ (slack 2), $x_C = 1 < 2$ (slack 1). Complementary slackness $y_i s_i = 0$ forces

$$y_3 = y_4 = y_5 = 0.$$

The positive primal variables $x_A, x_C > 0$ force their dual constraints to be tight ($x_j r_j = 0$):

$$y_1 - y_3 = 4, \quad y_1 + y_2 - y_5 = 5 \implies y_1 = 4, \quad y_2 = 1.$$

So $y^* = (4, 1, 0, 0, 0)$.

Verification. Dual feasibility: $y_1 - y_3 = 4 \leq 4$ (tight), $y_2 - y_4 = 1 \leq 3$ (slack $r_B = 2$), $y_1 + y_2 - y_5 = 5 \leq 5$ (tight), $y^* \geq 0$. Dual objective $2(4) + 1(1) = 9$. Complementary slackness, all ten products:

$$\begin{aligned} y_1 s_1 &= 4 \cdot 0, & y_2 s_2 &= 1 \cdot 0, & y_3 s_3 &= 0 \cdot 1, & y_4 s_4 &= 0 \cdot 2, & y_5 s_5 &= 0 \cdot 1; \\ x_A r_A &= 1 \cdot 0, & x_B r_B &= 0 \cdot 2, & x_C r_C &= 1 \cdot 0. \end{aligned}$$

All are zero. *Optimality:* x^* is primal feasible, y^* is dual feasible, and $c^\top x^* = 9 = b^\top y^*$. By weak duality no plan costs less than any dual value, so both are optimal (equivalently, by the complementary slackness theorem). The minimum cost is

$$C^* = 9\,000 \text{ MAD.}$$

Economically, $y_1 = 4$ says each day of tightening on the A–C path costs 4 (shorten A), and $y_2 = 1$ says the B–C deadline adds only 1 per day, because it is met “for free” by the day already taken off C.

Solution 3(e)

With $x^* = (1, 0, 1)$ the durations are A: 5 days, B: 5 days, C: 3 days. Schedule:

Activity	Start (day)	Duration	Finish (day)
A	0	5	5
B	0	5	5
C	5	3	8

C starts at $\max(5, 5) = 5$ and the project finishes at day 8, meeting the deadline. The cost is 9 000 MAD, so the actual saving relative to the manager’s plan of (c) is

$$11\,000 - 9\,000 = 2\,000 \text{ MAD,}$$

inside the interval $[0, 3\,000]$ predicted by the bounds.

Solution 3(f)

Yes, the optimal reduction plan is unique. Let x be *any* optimal primal solution. Since $y^* = (4, 1, 0, 0, 0)$ is optimal for the dual, the pair (x, y^*) is an optimal pair and must satisfy complementary slackness (the “only if” direction of the theorem). Therefore:

- $y_1^* = 4 > 0$ forces the first constraint to be tight: $x_A + x_C = 2$;
- $y_2^* = 1 > 0$ forces the second to be tight: $x_B + x_C = 1$;
- the dual constraint of x_B has slack $r_B = 3 - (y_2^* - y_4^*) = 2 > 0$, which forces $x_B = 0$.

From $x_B = 0$ and $x_B + x_C = 1$ we get $x_C = 1$, and then $x_A = 1$. Hence $x = (1, 0, 1) = x^*$: no other optimal plan exists.

(The same conclusion is visible in (d): $C(t)$ has a strict minimum at $t = 1$, and for fixed t any plan other than $(2 - t, \max(1 - t, 0), t)$ costs strictly more.)

Remark

In Problem 3 the manager’s plan was also feasible and only 2 000 MAD worse. Duality let us say, *before* solving, that it was at most 3 000 MAD from optimal, and then certify the exact optimum by exhibiting a matching dual solution. That is the practical content of Lecture 4: a dual solution is a proof.